

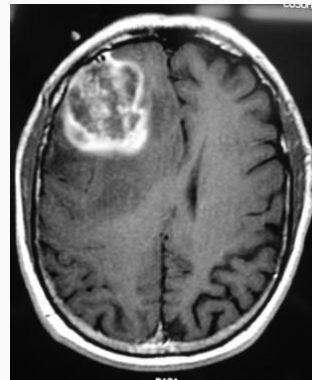
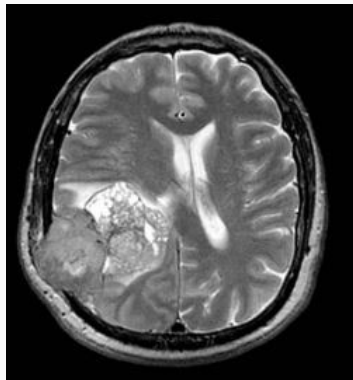
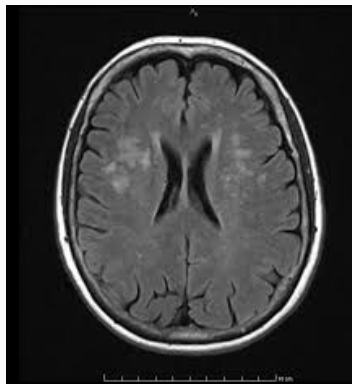
Neural Networks for Tumor Detection and Classification from MRI

Calvin Tyler



Refresher Summary

This project focused on using the ResNet architecture for image classification. A pretrained model was modified and retrained to detect and classify tumors from MRI scans.



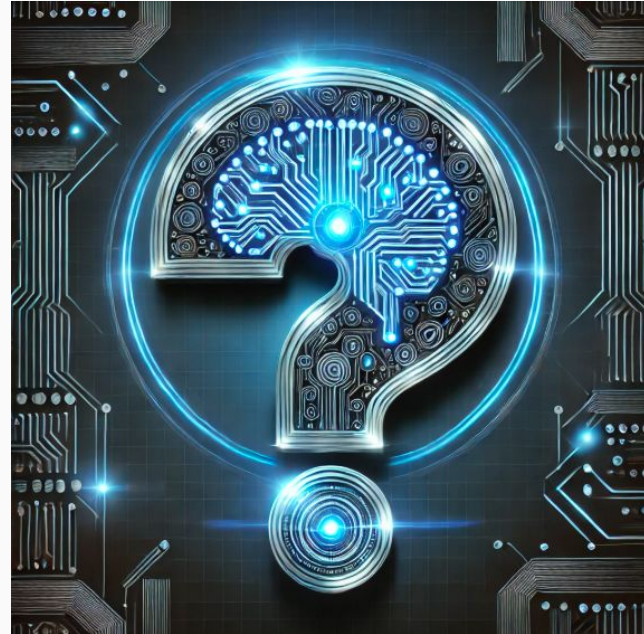
Progress Since Intermediary Update

- Model was functioning well
- Investigated interpretability
- Improved model
- Results analysis



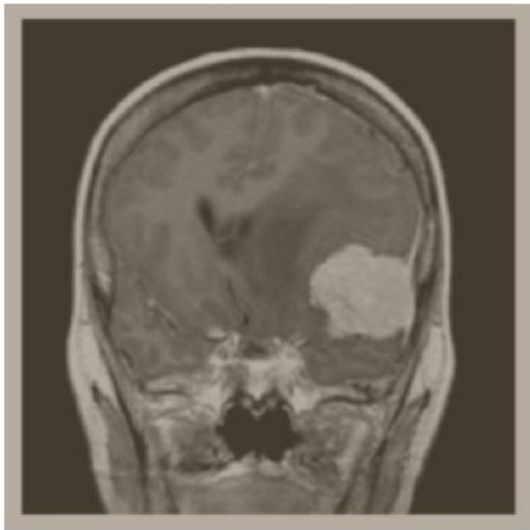
Explainability

- Grad-Cam
- Integrated Gradients
- Occlusion

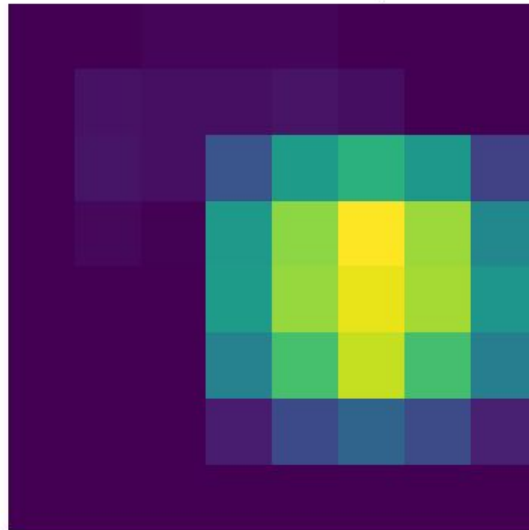


Grad-Cam

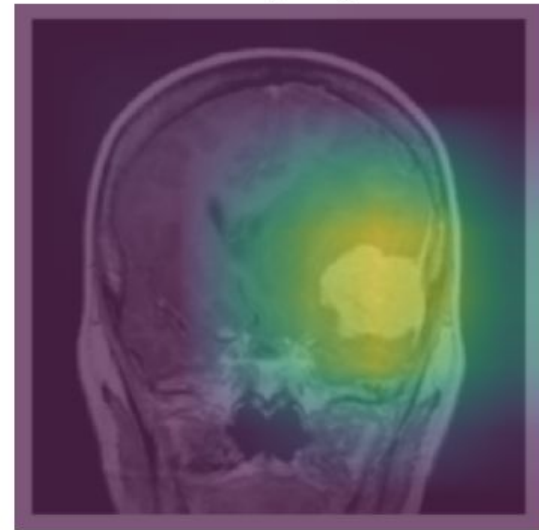
Original Image



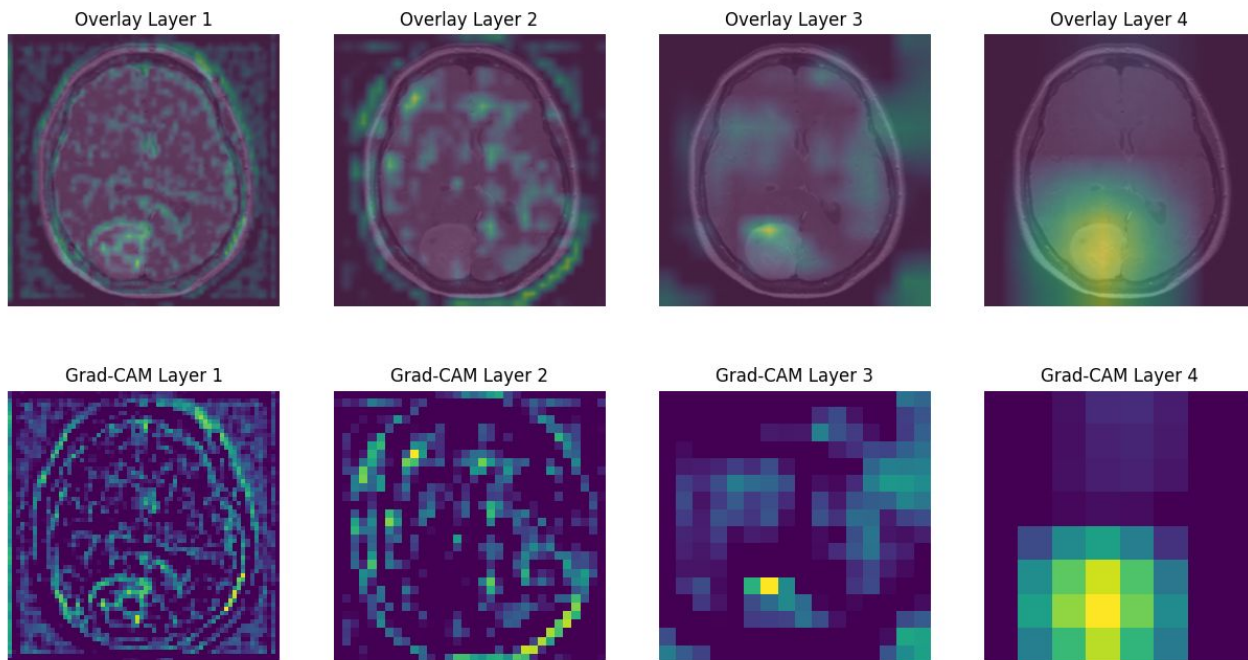
Grad-CAM Heatmap



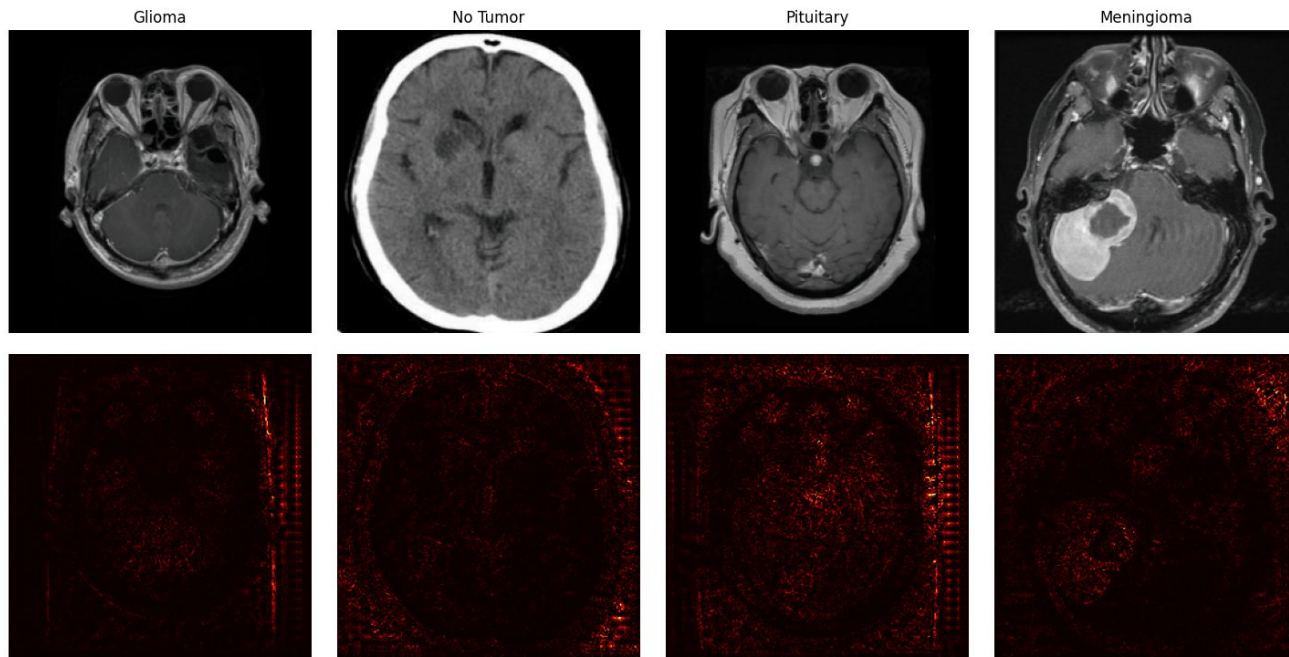
Overlay Image



Grad-Cam

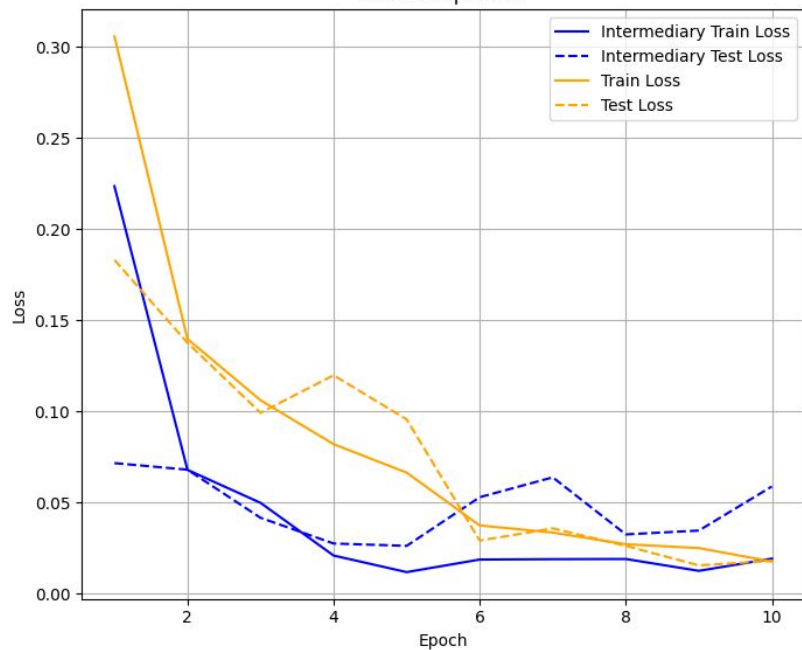


Integrated Gradients

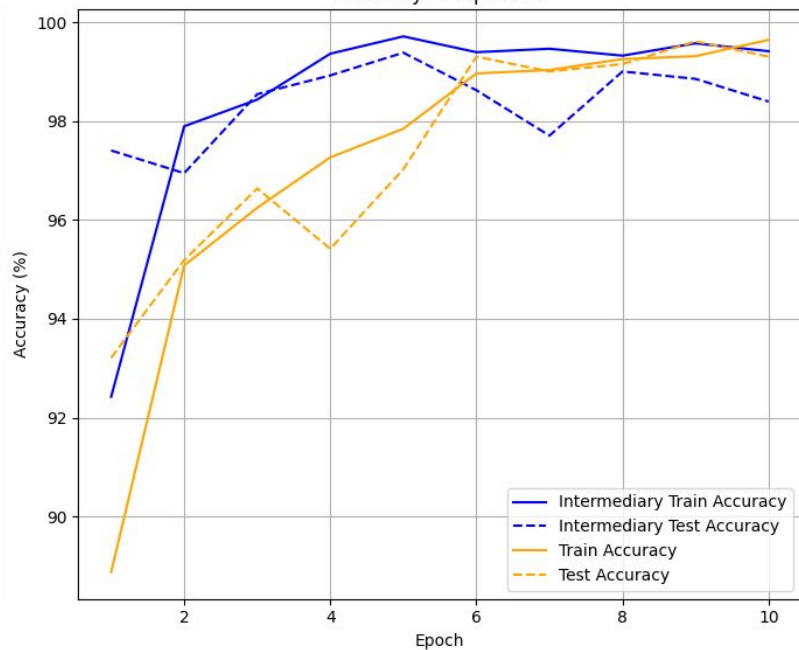


Results

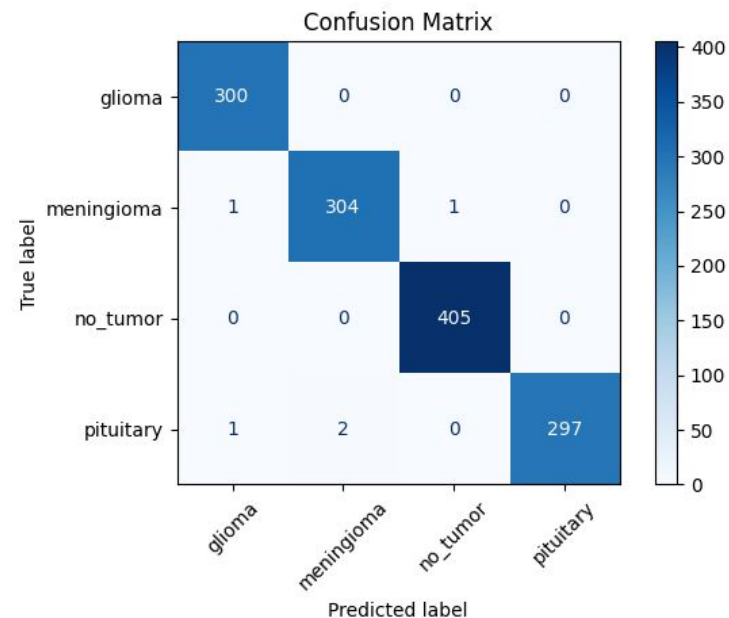
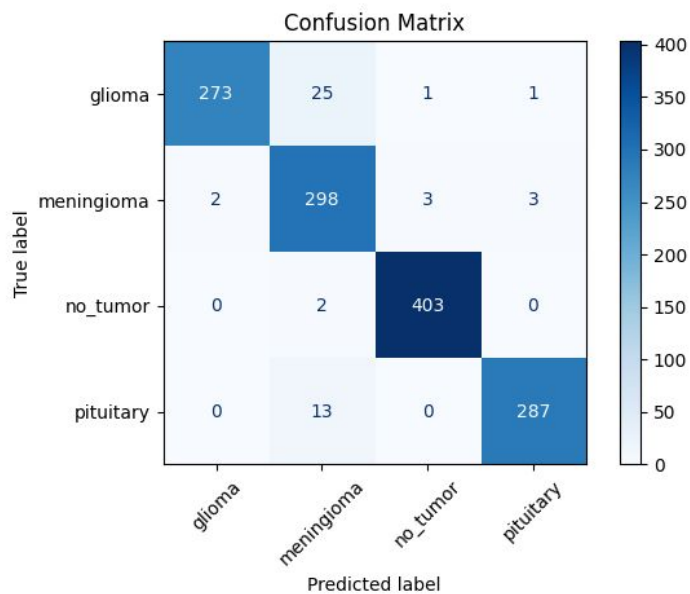
Loss Comparison



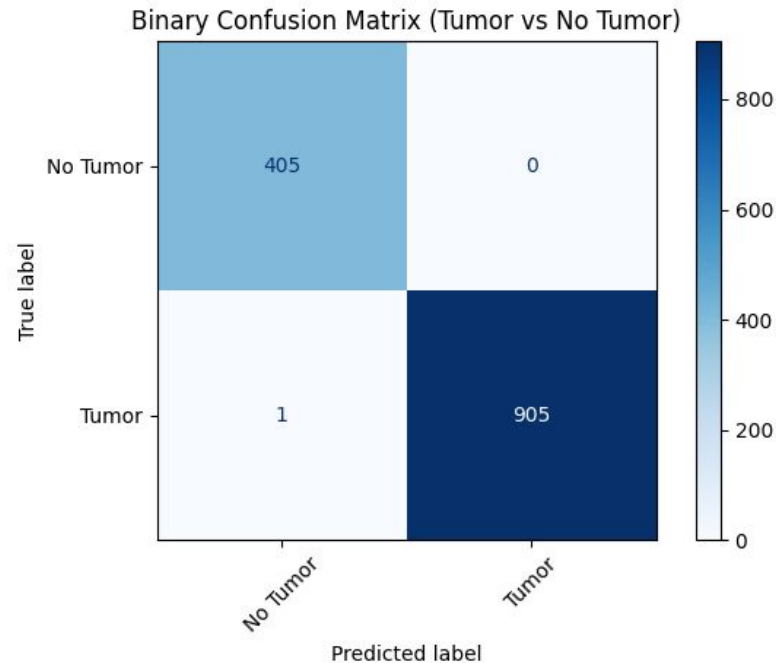
Accuracy Comparison



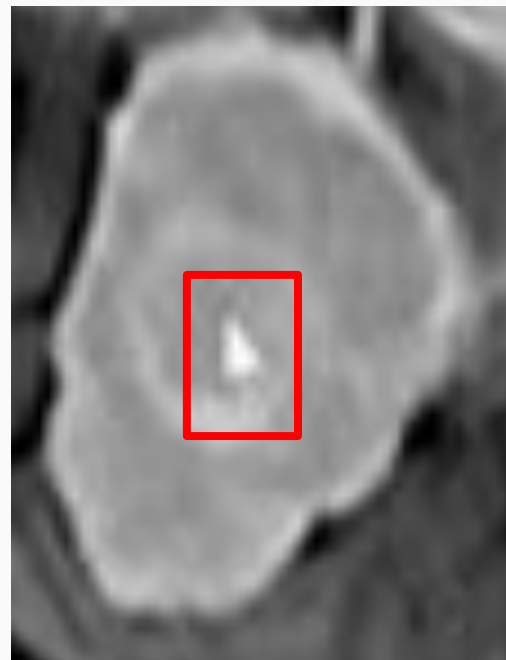
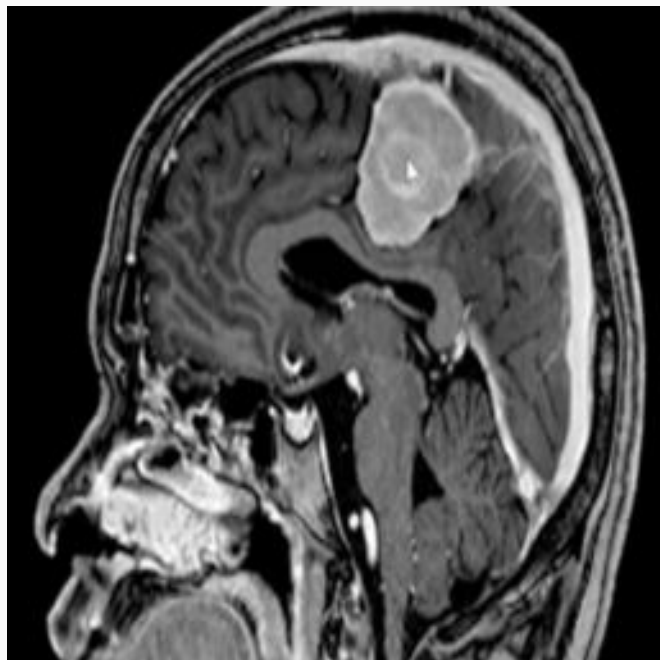
Confusion Matrix



Confusion Matrix

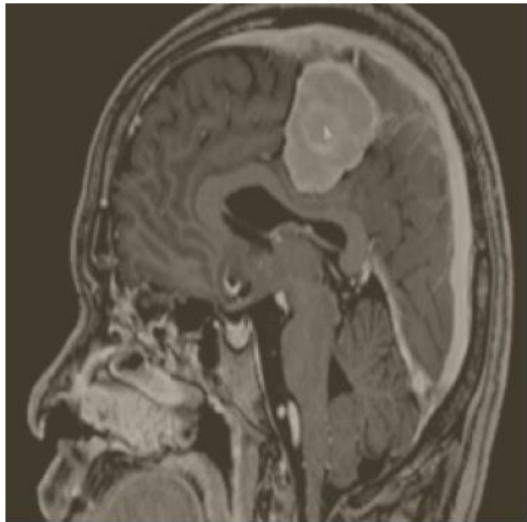


Mislabeled Data

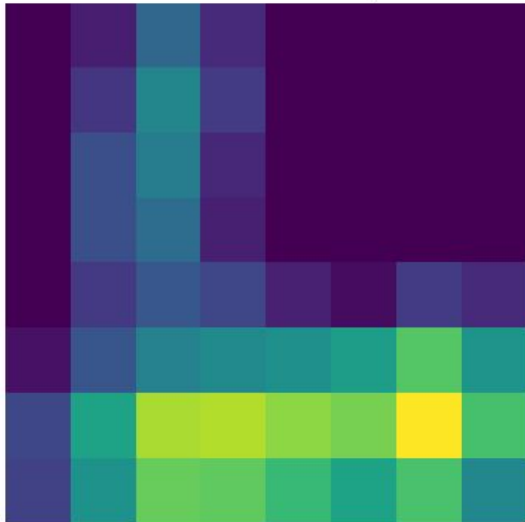


Mislabeled Data

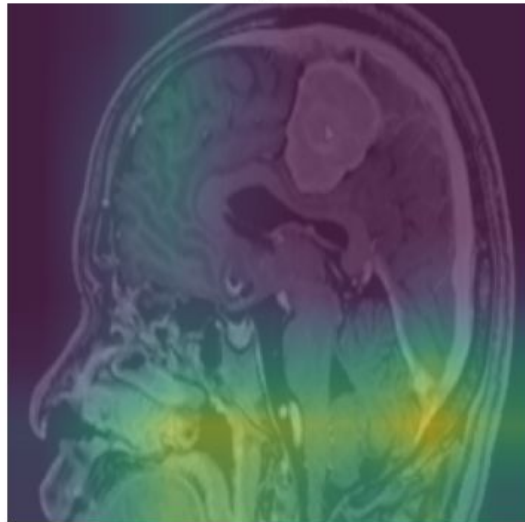
Original Image



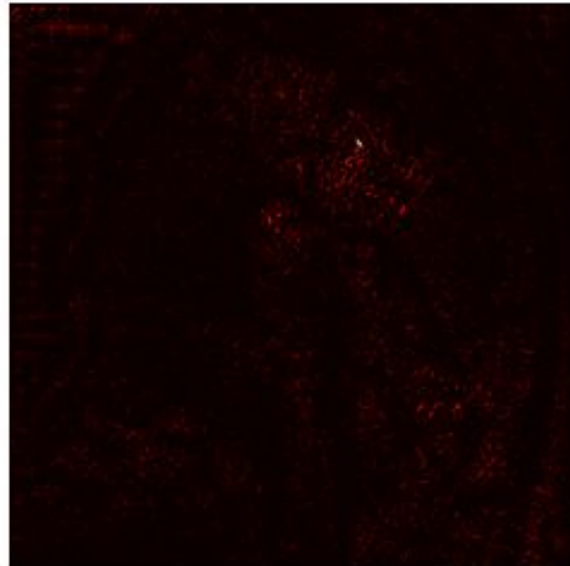
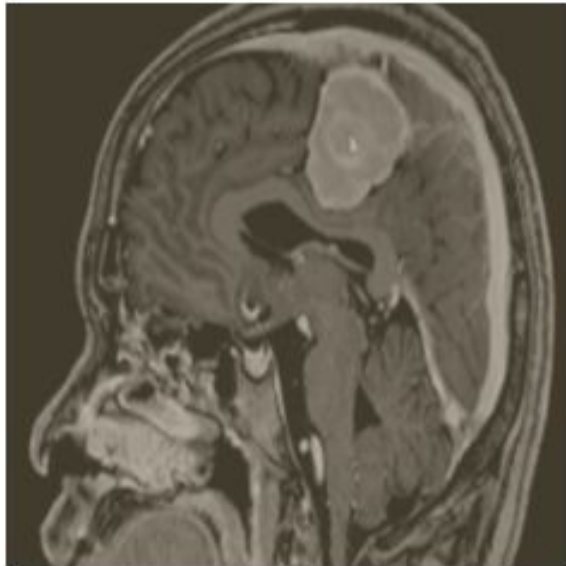
Grad-CAM Heatmap



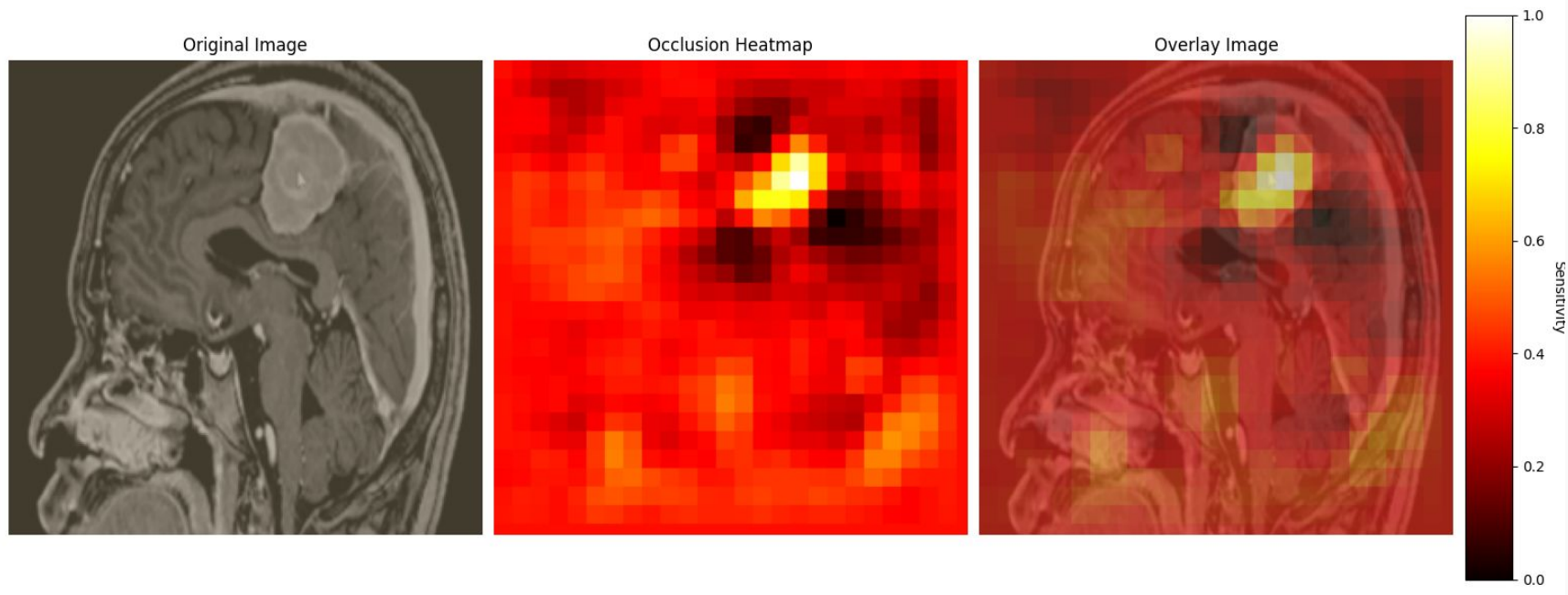
Overlay Image



Mislabeled Data



Mislabeled Data



Conclusion

- Initial scope accomplished
- Future exploration explored
- Explainability goals achieved

